What is the model actually looking at?

Making the self-supervised 4D-STEM "grain map" explainable in physical terms

A non-technical guide to the interpretability tests · IMC & NaPHI · Daniel Khaykelson

1. The setup, and the problem

Our model (a self-supervised network called DINO) looks at the electron diffraction pattern recorded at every probe position in a 4D-STEM scan and sorts each position into one of ∼14 groups—producing a "grain map" *with no labels and no chemistry input.* It works, but it is a black box: we do not know *which physical property* defines the groups (crystallinity? thickness? crystal orientation? something new?). The four tests below answer that, each from a different angle, so that together they triangulate a trustworthy physical explanation.

2. The four tests (plain idea + what it measures)

1. Probing — what the model pays attention to

Plain idea: if we can "read off" a known property (say crystallinity) from the model's internal notes for each point, then the model is keeping track of it.
What we measure: we fit a simple linear predictor from the model's 128-number fingerprint to each measured property; the score R^2 (0–1) says how strongly that property is encoded.

2. Ablation — what the model depends on

Plain idea: like covering part of a photo to see whether you can still recognise it.
What we measure: we blank out one ingredient of the diffraction pattern (overall brightness, fine angular detail, or inner vs. outer rings), re-run the model, and compare the new map to the original (agreement = ARI). If the map survives the edit, that ingredient was not needed; if it falls apart, the model depended on it.

3. Classical check — is the map actually new?

Plain idea: does the AI just redraw what a standard, decades-old analysis already shows, or something genuinely new?
What we measure: we cluster classical 4D-STEM features (a virtual dark-field image, the azimuthally-averaged profile, and PCA/NMF decompositions of the patterns) into the same number of groups, and measure how much they agree with the AI map (ARI/AMI).

4. Orientation check — is it just sorting by crystal direction?

Plain idea: a tilted crystal diffracts differently; maybe the model is only sorting by grain *orientation*.
What we measure: where a crystal structure (CIF) is available, we index the orientation at each point (ACOM) and measure agreement with the AI map. We also use Grad-CAM to see *which part* of the pattern the model looks at.

How to read the two numbers. R^2 — how predictable a property is from the model (0 = ignored $\rightarrow$ 1 = fully encoded). **ARI / AMI** — how much two maps agree, corrected for chance (0 = no better than random $\rightarrow$ 1 = identical maps).

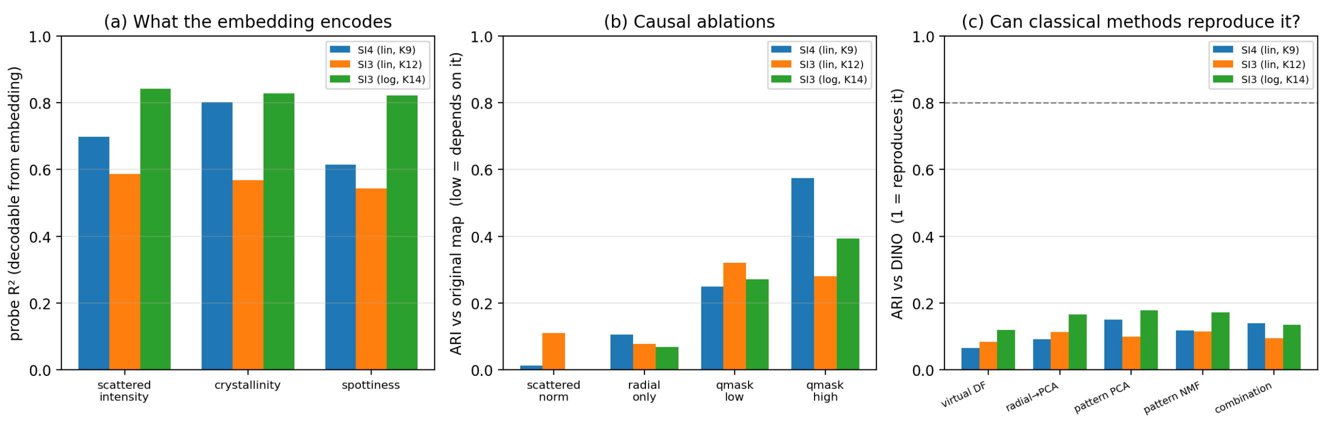
3. What we found

- The groups are driven by the **overall scattered (diffracted) intensity** together with the **inner, low-angle (low-q) 2-D diffraction structure** — the arrangement of the inner rings/spots ($q \lesssim 2.1\,\mathrm{nm}^{-1}$, i.e. $d \gtrsim 4.7\,\text{Å}$).
- It is **not** based on crystal orientation: agreement with the indexed zone axis is essentially zero (AMI ≤ 0.06), because the model is trained to be rotation-invariant by design.
- A 1-D radial profile alone is **not enough** — averaging away the angles collapses the map; the full 2-D pattern is needed.
- **Uniqueness is material-dependent.** For IMC, no classical method reproduces the map (best agreement ARI ≈ 0.15): the AI map is *distinctive*. For NaPHI, classical methods get about halfway (ARI ≈ 0.5): there the AI stays closer to a classical decomposition.
- Logarithmic intensity stretching (an image-prep choice) keeps the same physical basis but *sharpens* it — more, cleaner groups.

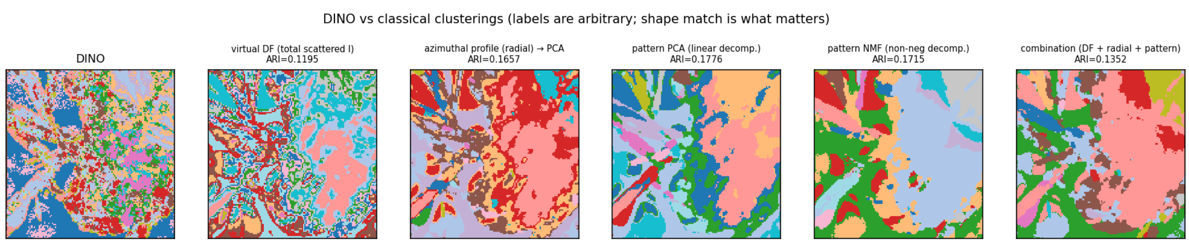
4. What it means (and one caveat)

The model groups regions by *how much, and what, they diffract at low angles* — crystallinity, molecular packing, and sample thickness. That is a physically sensible axis, not an arbitrary black box. For IMC it is genuinely distinctive (a standard virtual-detector or PCA map does not reproduce it); for NaPHI it stays closer to classical analysis.

Caveat: "scattered intensity" blends crystallinity with sample thickness — both raise the total diffracted signal. An independent thickness map (e.g. a t/λ estimate) would be needed to separate the two. Orientation/phase analysis is folded in automatically wherever a CIF is available; the conclusions above hold with or without it.



Cross-sample summary. **(a)** which properties are encoded (probe R^2); **(b)** causal ablations (low bar = the model depends on that signal); **(c)** how well classical methods reproduce the AI map (all ≤ 0.18 for IMC). Bars: SI4, SI3 (linear), SI3 (log-stretch).



IMC: the AI map (far left) vs. classical clusterings. Colours are arbitrary — what matters is the spatial shape. Classical methods catch the coarse regions but miss the fine grain structure the AI resolves.